Factorizarea LU

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & \dots & 0 \\ l_{21} & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{m1} & l_{m2} & \dots & l_{mm} \end{bmatrix} \cdot \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1m} \\ 0 & u_{22} & \dots & u_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{mm} \end{bmatrix}$$

$\boxed{A = L \cdot U}$, unde L = inferior triunghiulară ; U = superior triunghiula

Se disting trei factorizări:

I Factorizarea Crout

- condiție : $u_{ii} = 1$, pt $i = 1:m$ (U are 1 pe diag princ)

Pentru $p = 1:m$

$$\boxed{l_{ip} = a_{ip} - \sum_{k=1}^{p-1} l_{ik} \cdot u_{kp}}, \quad i = p:m$$

$$\boxed{u_{pj} = \frac{1}{l_{pp}} \left(a_{pj} - \sum_{k=1}^{p-1} l_{pk} \cdot u_{kj} \right)}, \quad j = p+1:m$$

II Factorizarea Doolittle

- condiție : $l_{ii} = 1$, pt $i = 1:m$ (L are 1 pe diag princ)

Pentru $p = 1:m$

$$\boxed{u_{pj} = a_{pj} - \sum_{k=1}^{p-1} l_{pk} \cdot u_{kj}} \quad j = p:m$$

$$\boxed{l_{ip} = \frac{1}{u_{pp}} \left(a_{ip} - \sum_{k=1}^{p-1} l_{ik} \cdot u_{kp} \right)}, \quad i = p+1:m$$

III Factorizarea Cholesky

- condiții : $U = L^T$; matricea A trebuie să fie simetrică ($A = A^T$) și pozitiv definită ($x^T A x > 0 ; \forall x \in \mathbb{R}^m ; x \neq 0$)

Pentru $i = 1:m$

Pentru $j = 1:i-1$

$$\boxed{l_{ij} = \frac{a_{ij} - \sum_{k=1}^{j-1} l_{ik} \cdot l_{jk}}{l_{jj}}}$$

$$\boxed{l_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} l_{ik}^2}}$$

L.

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Exemple

1. Determinați L și U din factorizarea Crout pentru matricea

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 5 & 1 \\ 2 & 1 & 2 \end{bmatrix}.$$

Rezolvare

$$\begin{bmatrix} 2 & -1 & 3 \\ 4 & 5 & 1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \cdot \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} l_{11} = 2 \\ l_{11}\, u_{12} = -1 \Rightarrow u_{12} = -1/2 \\ l_{11} \cdot u_{13} = 3 \Rightarrow u_{13} = 3/2 \\ l_{21} = 4 \\ l_{21} \cdot u_{12} + l_{22} = 5 \Rightarrow l_{22} = 5 - 4 \cdot \left(\frac{-1}{2}\right) = 7 \\ l_{21} \cdot u_{13} + l_{22} \cdot u_{23} = 1 \Rightarrow 7 \cdot u_{23} = 1 - 4 \cdot 3/2 \Rightarrow u_{23} = -5/7 \\ l_{31} = 2 \\ l_{31} \cdot u_{12} + l_{32} = 1 \Rightarrow l_{32} = 1 - 2 \cdot \left(-\frac{1}{2}\right) = 2 \\ l_{31} \cdot u_{13} + l_{32} \cdot u_{23} + l_{33} = 2 \Rightarrow l_{33} = 2 - 2\left(\frac{3}{2}\right) - 2 \cdot \left(-\frac{5}{7}\right) = \\ \qquad = -1 + \frac{10}{7} - \frac{9}{7} \end{cases}$$

$$\Rightarrow L = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 7 & 0 \\ 2 & 2 & 3/7 \end{bmatrix} \text{ și } U = \begin{bmatrix} 1 & -1/2 & 3/2 \\ 0 & 1 & -5/7 \\ 0 & 0 & 1 \end{bmatrix}$$

De reținut! $l_{i1} = a_{i1}\,;\ i = 1:n\,;\ u_{1j} = \dfrac{a_{1j}}{l_{11}}\,;\ j = 2:n$

2. Determinați L și U folosind Doolittle pentru matricea

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 5 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

Rezolvare

$$\begin{bmatrix} 2 & -1 & 3 \\ 4 & 5 & 1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \cdot \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

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$u_{11} = a_{11} = 2$

$l_{21} = \frac{1}{u_{11}}\left(a_{21}\right) = \frac{4}{2} = 2$

$u_{12} = a_{12} = -1$

$u_{13} = a_{13} = 3$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix} \qquad U = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 7 & -5 \\ 0 & 0 & \frac{3}{7} \end{bmatrix}$$

$u_{22} = a_{22} - \sum_{k=1}^{1} l_{2k} \cdot u_{k2} =$

$= a_{22} - l_{21} \cdot u_{12} = 5 - 2 \cdot (-1) = 7$

$u_{23} = a_{23} - \sum_{k=1}^{1} l_{2k} \cdot u_{k3}$

$= a_{23} - l_{21} \cdot u_{13} = 1 - 2 \cdot 3 = -5$

$u_{33} = a_{33} - \sum_{k=1}^{2} l_{3k} \cdot u_{k3}$

$= a_{33} - l_{31} \cdot u_{13} - l_{32} \cdot u_{23} = 2 - 1 \cdot 3 - \frac{2}{7} \cdot (-5) = -1 + \frac{10}{7} = \frac{3}{7}$

$l_{31} = \frac{1}{u_{11}}\left(a_{31}\right) = \frac{2}{2} = 1$

$l_{32} = \frac{1}{u_{22}}\left(a_{32} - \sum_{k=1}^{1} l_{3k} \cdot u_{k2}\right) = \frac{1}{u_{22}}\left(a_{32} - l_{31} \cdot u_{12}\right) =$

$= \frac{1}{7}\left(1 - 1 \cdot (-1)\right) = \frac{2}{7}$

Rezolvarea sistemelor triunghiulare

- Sistemul inferior triunghiular (SIT)

Forma:
$$\begin{cases} a_{11}x_1 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \\ \vdots \quad \vdots \quad \ddots \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mm}x_m = b_m \end{cases}$$

Solutia: $\boxed{x_i = \frac{b_i - \sum_{j=1}^{i-1} A_{ij} \cdot x_j}{A_{ii}}}$: pt $i = 1 : m$

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- Sistemul superior triunghiular (SST)

Forma:
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m = b_1 \\ a_{22}x_2 + \dots + a_{2m}x_m = b_2 \\ \vdots \\ a_{mm}x_m = b_m \end{cases}$$

Soluția: $\boxed{x_i = \dfrac{b_i - \sum_{j=i+1}^{m} A_{ij} \cdot x_j}{A_{ii}}}$; pt $i = 1:m$

Dacă trebuie rezolvat $Ax = b$; $A = L \cdot U \Rightarrow L \cdot U \cdot x = b$

Se notează $U \cdot x \overset{not}{=} y \xrightarrow{SST} \dots \Rightarrow y = \begin{bmatrix} \vdots \end{bmatrix} \Rightarrow$

$\Rightarrow L \cdot y = b \xrightarrow{SIT} \dots \Rightarrow b = \begin{bmatrix} \vdots \end{bmatrix}$

Funcții OCTAVE utile

```
function [L U] = crout(A)
  [m m] = size(A)
  L = zeros(m)
  U = eye(m)
  L(1:m,1) = A(1:m,1)
  for p = 1:m
    for i = p:m
      s = 0
      for k = 1:p-1
        s = s + L(i,k)*U(k,p)
      endfor
      L(i,p) = A(i,p) - s
    endfor
    for j = p+1:m
      s = 0
      for k = 1:p-1
        s = s + L(i,k)*U(k,j)
      endfor
      U(p,j) = (A(p,j) - s)/L(p,p)
    endfor
  endfor
```

```
function [L U] = doolittle(A)
  [m,m] = size(A)
  L = eye(m)
  U = zeros(m)
  for p = 1:m
    for j = p:m
      s = 0
      for k = 1:p-1
        s = s + L(p,k)*U(k,j)
      endfor
      U(p,j) = A(p,j) - s
    endfor
    for i = p+1:m
      s = 0
      for k = 1:p-1
        s = s + L(i,k)*U(k,p)
      endfor
      L(i,p) = (A(i,p) - s)/U(p,p)
    endfor
  endfor
endfunction
```

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Transformări ortogonale. Householder și Givens

Noțiuni teoretice

Fie $x, y \in \mathbb{R}^m$, $x = [x_1\ x_2 \dots x_m]^T$ și $y = [y_1\ y_2 \dots y_m]^T$.

- produs scalar: $\langle x, y \rangle = \sum_{i=1}^{m} x_i y_i = y^T x$
- norma euclidiană: $\|x\|_2 = \sqrt{\langle x, x \rangle} = \sqrt{x^T x} = \sqrt{\sum_{i=1}^{m} x_i^2}$
- vectori ortogonali $\Rightarrow \langle x, y \rangle = 0$
- vectori ortonormați $\Rightarrow$ sunt ortogonali. $\|x\|_2 = 1$ și $\|y\|_2 = 1$

$H \in \mathbb{R}^{m \times m}$ este ortogonală dacă are coloanele vectori ortonormați. Proprietăți:

1. $H^T \cdot H = H \cdot H^T = I_m$
2. $H^{-1} = H^T$
3. $\|Hx\|_2 = \|x\|_2$
4. $\|H\|_2 = 1$
5. $\|HA\|_2 = \|A\|_2$
6. $\det(H) = \pm 1$

Transformarea Householder

Pentru $A \in \mathbb{R}^{m \times m}$, definim Householder $R = H \cdot A$ folosind reflectorii H_p, a.î.

$$H = H_{\min(m-1, m)} \dots H_p \dots H_2 H_1$$

$$\boxed{H_p = I_m - 2 \frac{v_p \cdot v_p^T}{v_p^T \cdot v_p}}, \text{ unde:}$$

- $v_p = [0\ 0 \dots v_{pp} \dots v_{mp}]^T \rightarrow$ vector Householder
- $v_{pp} = a_{pp} + \sigma_p$
- $v_{ip} = a_{ip}\ ; \forall i > p$
- $\sigma_p = sign(a_{pp}) \sqrt{\sum_{i=p}^{m} a_{ip}^2}$
- $\beta_p = \sigma_p v_{pp}$
- $a_p = [a_{1p}\ a_{2p} \dots a_{pp} \dots a_{mp}]^T \rightarrow$ coloana p din A.

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La înmulțirea $A = H_p A$ efectuăm:

- coloanele $1:p-1$ din matricea A rămân neschimbate
- coloana p din matricea A se modifică astfel:
  - elementele de pe liniile $1:p-1$ rămân neschimbate
  - $a_{pp} = -\sigma_p$
  - $a_{ip} = 0,\ \forall i > p$
- coloanele $j = p+1:n$ din matricea A se modifică:
  - elementele de pe liniile $1:p-1$ rămân neschimbate
  - $a_{ij} = a_{ij} - \gamma_j \cdot v_{ip};\ \forall i > p;\ \gamma_j = \frac{\sum_{i=p}^{m} v_{ip} \cdot a_{ij}}{\beta_p}$

Exemplu

1. Determinați factorizarea QR folosind transf. Householder pentru matricea $A = \begin{bmatrix} 2 & 4 & 5 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

Rezolvare

Pasul $p=1$: $A_1 \overset{not}{=} A;\ a_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix};\ \sigma_1 = \text{sign}(a_{11}) \|a_1\|_2 =$

$= +\sqrt{2^2+1^2+2^2} = \sqrt{9} = 3;$

$v_1 = \begin{bmatrix} a_{11} + \sigma_1 \\ a_{12} \\ a_{13} \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \Rightarrow H_1 = I_3 - 2 \cdot \frac{v_1 \cdot v_1^T}{\|v_1\|^2} \Rightarrow$

$\Rightarrow H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{\sqrt{5^2+2^2+1^2}} \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 5 & 1 & 2 \end{bmatrix} =$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{\sqrt{30}} \begin{bmatrix} 25 & 5 & 10 \\ 5 & 1 & 2 \\ 10 & 2 & 4 \end{bmatrix} =$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{15} \begin{bmatrix} 25 & 5 & 10 \\ 5 & 1 & 2 \\ 10 & 2 & 4 \end{bmatrix} =$

$= \begin{bmatrix} -2/3 & -1/3 & -2/3 \\ -1/3 & 14/15 & -2/15 \\ -2/3 & -2/15 & 11/15 \end{bmatrix} \Big| \cdot A_1$

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$$A_2 \overset{not}{=} H_1 \cdot A_1 = -\frac{1}{15}\begin{bmatrix} 10 & 5 & 10 \\ 5 & -14 & 2 \\ 10 & 2 & -11 \end{bmatrix}\begin{bmatrix} 2 & 4 & 5 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix} =$$

$$= -\frac{1}{15}\begin{bmatrix} 45 & 45 & 45 \\ 0 & 36 & 9 \\ 0 & 27 & 63 \end{bmatrix} = \begin{bmatrix} -3 & -3 & -3 \\ 0 & -12/5 & -3/5 \\ 0 & -9/5 & -21/5 \end{bmatrix}$$

Obs Luăm submatricea din colțul din dreapta jos pentru pasul al doilea

Pasul $p=2$ $\quad \tilde{A}_2 = \begin{bmatrix} -12/5 & -3/5 \\ -9/5 & -2/5 \end{bmatrix}$

$\tilde{a}_1$

$$\tilde{a}_1 = \begin{bmatrix} -\frac{12}{5} \\ -\frac{9}{5} \end{bmatrix}; \quad \sigma_2 = \text{sign}(\tilde{a}_{11}) \cdot \|\tilde{a}_1\|_2 = -\sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{9}{5}\right)^2} =$$

$$= -\sqrt{\frac{144+81}{25}} = -\sqrt{\frac{225}{25}} = -\sqrt{9} = -3$$

$$v_2 = \begin{bmatrix} -\frac{12}{5} - 3 \\ -\frac{9}{5} \end{bmatrix} = \begin{bmatrix} -27/5 \\ -9/5 \end{bmatrix} \Rightarrow$$

$$\Rightarrow H_2 : I_2 - 2 \cdot \frac{v_2 \cdot v_2^T}{\|v_2\|^2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{2}{\frac{27^2+9^2}{25}}\begin{bmatrix} -27/5 \\ -9/5 \end{bmatrix}\begin{bmatrix} -\frac{27}{5} & -\frac{9}{5} \end{bmatrix} =$$

$$= \frac{50^2}{9^2(3^2+1)} \cdot \left(-\frac{9}{5}\right)\cdot\left(-\frac{9}{5}\right)\begin{bmatrix} 3 \\ 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix} \Big| \cdot A_2$$

$$H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 9/5 & 3/5 \\ 0 & 3/5 & 1/5 \end{bmatrix} \Big| \cdot A_2$$

$$\Rightarrow H_2 A_2 = \frac{-1}{25}\begin{bmatrix} 5 & 0 & 0 \\ 0 & 9 & 3 \\ 0 & 3 & 1 \end{bmatrix}\begin{bmatrix} -15 & -15 & -15 \\ 0 & -12 & -3 \\ 0 & 9 & 21 \end{bmatrix} =$$

$$= -\frac{1}{25}\begin{bmatrix} 75 & 75 & 75 \\ 0 & 135 & 90 \\ 0 & \cdots & \end{bmatrix} = R$$ (ar trebui să dea o matrice superior triunghiulară)

$$Q = (H_2 \cdot H_1)^T = H_1 \cdot H_2.$$

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Transformarea Givens

Folosim Givens pt a descompune $A \in \mathbb{R}^{m \times n}$ în $A = G^T \cdot R$, unde $G = G_{m-1,m}\, G_{m-2,m}\, G_{m-2,m-1} \dots G_{1m} \dots G_{13} G_{12}$, $R = GA$

O matrice de rotație Givens G_{kl} are forma

$$G_{kl} = \begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \cos\theta & \dots & -\sin\theta & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \sin\theta & \dots & \cos\theta & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$\rho = \sqrt{A(k,k)^2 + A(l,k)^2}$$

$$s = \sin\theta = -\frac{A(l,k)}{\rho} \; ; \; c = \cos\theta = \frac{A(k,k)}{\rho}$$

Înmulțirea $G_{kl} \cdot x$, unde x e un vector coloană, se efectuează astfel:

- $x(k) = c \cdot x(k) - s \cdot x(l)$
- $x(l) = s \cdot x(k) + c \cdot x(l)$
- restul elementelor rămân neschimbate.

Exemplu

1. Determinați factorizarea QR folosind transf. Givens pentru matricea $A = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix}$.

Rezolvare

- Anulăm $A^{(0)}(2,1) = 0$, $A^{(1)} = A$.

$$\rho = \sqrt{A^{(0)}(1,1)^2 + A^{(0)}(2,1)^2} = \sqrt{3^2 + 0^2} = 3$$

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$s = \sin\theta = -\frac{A^{(0)}(2,1)}{3} = -\frac{-0}{3} = 0$

$c = \cos\theta = \frac{A^{(0)}(1,1)}{3} = \frac{3}{3} = 1$

$G_{12}^* = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$; $G_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

se schimbă L_1 și L_2 din matricea $A^{(0)}$

$G_{12}^* \cdot \underbrace{\begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \end{bmatrix}}_{\text{submatrice a lui } A^{(0)}} \begin{matrix} L_1 \\ L_2 \end{matrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \end{bmatrix}$

$\Rightarrow A^{(1)} = G_{12} \cdot A^{(0)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix}$

- Anulăm $A^{(1)}(3,1) = 4$

$p = \sqrt{A^{(1)}(1,1)^2 + A^{(1)}(3,1)^2} = \sqrt{3^2+4^2} = 5$

$s = \sin\theta = -\frac{A^{(1)}(3,1)}{5} = -\frac{4}{5}$

$c = \cos\theta = \frac{A(1,1)}{5} = \frac{3}{5}$

$\Rightarrow G_{13} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3/5 & 4/5 \\ 0 & -4/5 & 3/5 \end{bmatrix}$

$\Rightarrow A^{(2)} = G_{13} A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3/5 & 4/5 \\ 0 & -4/5 & 3/5 \end{bmatrix} \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix} = \begin{bmatrix} \dots \end{bmatrix}$

Morala (coparenta): A ⇝ puține zerouri ⇒ Householder; multe zerouri ⇒ Givens

Algoritmul Gram-Schmidt

Considerăm $A = Q \cdot R$

$$[a_1\ a_2 \dots a_m] = [q_1\ q_2 \cdots q_m] \cdot \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1m} \\ 0 & r_{22} & \dots & r_{2m} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & r_{mm} \end{bmatrix}$$

- a_i este coloana i din matricea A
- q_i este coloana i din matricea Q
- r_{ij} ; $i \le j$ reprezintă elem din matricea R. superior triunghiulară.

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- Algoritm:

Pentru $j=1:n$

$r_{ij} = q_i^T a_j \; ; \; i = 1:j-1$

$aux = a_j - \sum_{i=1}^{j-1} r_{ij} q_i$

$r_{ij} = \|aux\|_2$

$q_i = \frac{aux}{r_{ij}}$

Algoritmul Gram-Schmidt modificat.

Pentru $i = 1:n$

$r_{ii} = \|a_i\|_2$

$q_i = \frac{a_i}{r_{ii}}$

Pentru $j = i+1 : n$

$r_{ij} = q_i^T \cdot a_j$

$a_j = a_j - q_i r_{ij}$

Exemplu

1. Determinați factorizarea QR folosind algoritmul Gram-Schmidt modificat pentru matricea $A = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix}$ și normal

Rezolvare (normal)

pas1 pas2 pas3

$$A = \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix} = [q_1 \; q_2 \; q_3] \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{bmatrix}$$

$a_1 \; a_2 \; a_3$

Pasul 1: $r_{11} = \|a_1\|_2 = \sqrt{3^2+0^2+4^2} = 5$

$q_1 = \frac{a_1}{r_{11}} = \begin{bmatrix} 3/5 \\ 0 \\ 4/5 \end{bmatrix}$

Pasul 2: $r_{12} = \langle q_1, a_2 \rangle = \frac{-6}{5} + 0 + \frac{16}{5} = 2$

$a_2 = a_2 - r_{12} \cdot q_1 = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix} - 2 \cdot \begin{bmatrix} 3/5 \\ 0 \\ 4/5 \end{bmatrix} = \begin{bmatrix} -16/5 \\ 3 \\ 12/5 \end{bmatrix}$

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$\Rightarrow r_{22} = \|u_2\| = \sqrt{\frac{256}{25} + 9 + \frac{144}{25}} = \sqrt{16+9} = 5$

$q_2 = \frac{u_2}{r_{22}} = \frac{1}{5}\begin{bmatrix} -16/5 \\ 3 \\ 12/5 \end{bmatrix}$

Pasul 3 $\quad r_{13} = \langle q_1, a_3 \rangle = \frac{3}{5}\cdot 3 + \frac{4}{5}\cdot 4 = 5$

$r_{23} = \langle q_2, a_3 \rangle = -\frac{16}{25}\cdot 3 + \frac{3}{5}\cdot 5 + \frac{12}{25}\cdot 4 = 3$

$u_3 = a_3 - r_{13}q_1 - r_{23}q_2 = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix} - 5\begin{bmatrix} 3/5 \\ 0 \\ 4/5 \end{bmatrix} - 3\begin{bmatrix} -16/25 \\ 3/5 \\ 12/25 \end{bmatrix} =$

$= \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} - \begin{bmatrix} -48/25 \\ 9/5 \\ 36/25 \end{bmatrix} = \begin{bmatrix} 48/25 \\ 16/5 \\ -36/25 \end{bmatrix}$

$\Rightarrow r_{33} = \|u_3\| = \sqrt{\frac{48^2}{25^2} + \frac{16^2}{5^2} + \frac{36^2}{25^2}} = \sqrt{\frac{6^2(8^2+6^2)}{25^2} + \frac{80^2}{25^2}} =$

$= \sqrt{\frac{6^2\cdot 10^2 + 80^2}{25^2}} = \sqrt{\frac{10^2(6^2+8^2)}{25^2}} = \sqrt{\frac{10^2\cdot 10^2}{25^2}} = \frac{100}{25} = 4$

$q_3 = \frac{u_3}{r_{33}} = \frac{1}{4}\begin{bmatrix} 48/25 \\ 16/5 \\ -36/25 \end{bmatrix} = \begin{bmatrix} 12/25 \\ 4/5 \\ -9/25 \end{bmatrix}$

- modificat

Pasul 1 ($i=1$) $\quad r_{11} = \|a_1\|_2 = \sqrt{3^2+4^2} = 5$

$q_1 = \frac{a_1}{r_{11}} = \frac{1}{5}\begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$

$r_{12} = \langle q_1; a_2 \rangle = 2$

$a_2 = a_2 - q_1 \cdot r_{12} = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix} - \frac{2}{5}\begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} -16/5 \\ 3 \\ 12/5 \end{bmatrix}$

$r_{13} = \langle q_1; a_3 \rangle = 5$

$a_3 = a_3 - q_1 r_{13} = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$

Pasul 2 ($i=2$) $\quad r_{22} = \|a_2\|_2 = 5$

$q_2 = \frac{a_2}{r_{22}} = \frac{1}{5}\begin{bmatrix} -16/5 \\ 3 \\ 12/5 \end{bmatrix}$

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$r_{23} = \langle q_2, a_3 \rangle = 3$

$$a_3 = a_3 - q_2 \cdot r_{23} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} - \frac{3}{5}\begin{bmatrix} -16/5 \\ 12/5 \end{bmatrix} = \begin{bmatrix} +48/25 \\ 16/5 \\ -36/25 \end{bmatrix}$$

Pasul 3 $(i=3)$ $r_{33} = \|a_3\|_2 = 4$

$$q_3 = \frac{a_3}{r_{33}} = \begin{bmatrix} 12/25 \\ 4/5 \\ -9/25 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & -2 & 3 \\ 0 & 3 & 5 \\ 4 & 4 & 4 \end{bmatrix} = \underbrace{\begin{bmatrix} 3/5 & -16/25 & 12/25 \\ 0 & 3/5 & 4/5 \\ 4/5 & 12/25 & -9/25 \end{bmatrix}}_{Q} \cdot \underbrace{\begin{bmatrix} 5 & 2 & 5 \\ 0 & 5 & 3 \\ 0 & 0 & 4 \end{bmatrix}}_{R}$$

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